A coloring-free approach to materialising Jacobians and Hessians in JAX by leveraging sparsity

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Abstract

Sparse Hessian and Jacobian extraction is a bottleneck for second-order methods on large structured problems: dense AD wastes work on zeros, while coloring schemes (e.g. asdex) recover only a conservative over-approximation of the true sparsity pattern and pay for it in compressed evaluations. We present lineaxpr, a JAX transformation that materialises exact Jacobians and Hessians by performing a single structural walk over the linear jaxpr produced by jax.linearize. Each primitive is handled by a small rule that propagates a mixed-format linear-operator state (dense, BCOO, and a banded ELLPACK form, BEllpack) without densifying intermediates; identity-seeded forward propagation then yields the Jacobian directly, and composing two passes yields the Hessian. The resulting matrices are bit-exact against vmap(hvp)(eye) references, and the BEllpack form makes shape-only operations such as slice, pad, and reshape free at the structural level. On a sweep of 275 unconstrained optimisation problems from the CUTEst/sif2jax suite, lineaxpr produces strictly tighter sparsity than coloring on every problem and matches or beats both jax.hessian and asdex on wall-clock time for the majority, with the largest wins on banded and block-banded structures where BEllpack avoids the densification step that both baselines incur. We expose a drop-in API (hessian, jacfwd, jacrev, plus bcoo variants) matching jax.hessian/jacfwd/jacrev.

1 Introduction

2 State of the art

3 Sparse representations

4 You only linearize once

5

$$\text{Identity} \to \text{Constant Diagonal} \to \text{Diagonal} \to \text{BEllpack} \to \text{BCOO}$$

6 Partial separability and the limits of coloring

Many of the CUTEst problems involve objective functions of the form:

$$f(x) = \sum_i f_i(x), \tag{1}$$

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Operation	Dense	values	in_cols	start	end
(seed) $x = \texttt{Identity}(5)$	$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$	$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$	0	5
$a = c \odot x,\, c = (3, 5, 7, 9, 11)$	$\begin{pmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 & 11 \end{pmatrix}$	$\begin{bmatrix} 3 \\ 5 \\ 7 \\ 9 \\ 11 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$	0	5
$b = a_{1:5}$	$\begin{pmatrix} 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 & 11 \end{pmatrix}$	$\begin{bmatrix} 5 \\ 7 \\ 9 \\ 11 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$	0	4
$d = a_{0:4}$	$\begin{pmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 \end{pmatrix}$	$\begin{bmatrix} 3 \\ 5 \\ 7 \\ 9 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$	0	4
$e = b - d$	$\begin{pmatrix} -3 & 5 & 0 & 0 & 0 \\ 0 & -5 & 7 & 0 & 0 \\ 0 & 0 & -7 & 9 & 0 \\ 0 & 0 & 0 & -9 & 11 \end{pmatrix}$	$\begin{bmatrix} -3 & 5 \\ -5 & 7 \\ -7 & 9 \\ -9 & 11 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$	0	4
$f = \texttt{pad}(e,\, 1 \to 0)$	$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ -3 & 5 & 0 & 0 & 0 \\ 0 & -5 & 7 & 0 & 0 \\ 0 & 0 & -7 & 9 & 0 \\ 0 & 0 & 0 & -9 & 11 \end{pmatrix}$	$\begin{bmatrix} -3 & 5 \\ -5 & 7 \\ -7 & 9 \\ -9 & 11 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$	1	5

Table 1: Walk for $f(x) = \mathrm{pad}\big((c \odot x)_{1:} - (c \odot x)_{:-1}\big)$ with $c = (3, 5, 7, 9, 11)$, $\mathrm{len}(x) = 5$. The mathematical Jacobian (parentheses, right-aligned for sign clarity) is contrasted with the BEllpack representation (square brackets, the literal in-memory tensors). Elementwise scaling rewrites `values` in place; slicing shifts the columnfband; subtraction concatenates two k=1 bands into a k=2 band; pad updates only `start`/`end`.

Table 2: Categorised list of linear JAX primitives supported by Ellpax (trailing `_p` dropped).

Group	Primitive
Shape-preserving unary	`conj, copy, imag, neg, real, cumsum, convert_element_type`[1]
Shape-affecting unary	`pad, reshape, broadcast_in_dim, reduce_sum, squeeze`
Indexing	`slice, rev, gather, transpose, scatter_add`
Shape-preserving multilinear	`add`[2]`, sub, mul, div`
Shape-affecting multilinear	`dot_general, concatenate`
Control flow	`cond, select_n, (jit)`
Multi-output unary	`split`

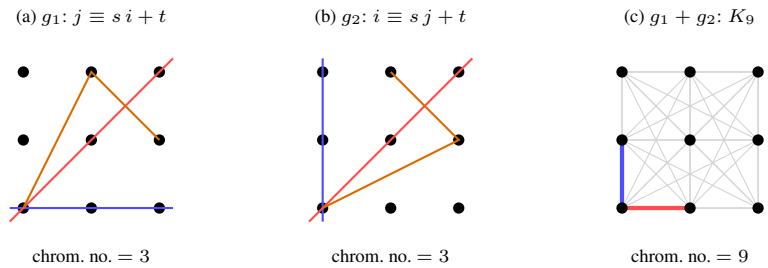

Figure 1: Each summand admits a valid 3-coloring of its column-conflict graph (columns sharing i never co-occur in a g_1-line; similarly j for g_2). Their union makes *every* pair of grid points share a line — the column conflict graph becomes K_9, requiring 9 colors. The highlighted edges in (c) show two pairs each covered by exactly one summand.

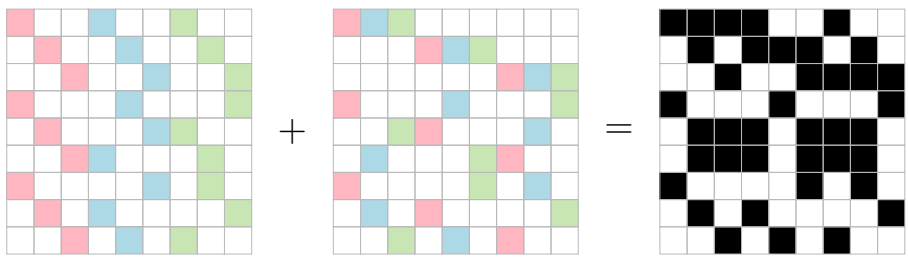

Figure 2: Boolean Jacobian patterns at $P = 3$. Each summand is 9×9 with 3 nonzeros per row; the column-conflict graph of $J(g_1)$ admits a 3-colouring grouped by row index i (panel a, columns of grid points sharing i never co-occur in a g_1-line), and likewise $J(g_2)$ admits a 3-colouring by j (panel b, the transverse partition). When the two Jacobians are added, every pair of distinct columns is joined by some g_1- or g_2-line, so the aggregate column-conflict graph is complete and no compression below 9 colours exists (panel c, shown without colouring).

where

7 Bib test

This section exercises the bibliography wiring; it should be removed before submission.

Single citation: foundational sparse Jacobian estimation [7]. Two-cite: graph-theoretic formulation [4, 5]. Three consecutive (compressed): coloring lineage [7, 18, 4]. Survey reference [11]; recent comparator work [15, 8].

Partial-separability [13, 6]; multifrontal-assembly analogy [9, 16]. Experiments on the CUTEst sweep [12, 2], bit-exact against the Pearlmutter HVP oracle [17]. Background AD and sparse storage [14, 1, 19]; host system [10, 3].

8 Conclusions

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A Technical appendices and supplementary material

Technical appendices with additional results, figures, graphs, and proofs may be submitted with the paper submission before the full submission deadline (see above). You can upload a ZIP file for videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the technical appendices.

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B Action of primitive rules on a BEllpack operand

We use the 5-tuple shorthand $\mathrm{BE}[V,\, C,\, s,\, e,\, T]$ for a BEllpack with values V, in-column tuple C of length k, row range $[s, e)$, and transpose flag T. The remaining fields (`out_size`, `in_size`, `batch_shape`) pass through unchanged unless explicitly modified. Each row below is the canonical structural case implemented in the current source; rules with several distinct branches are listed once per branch with a labelled left-hand side. When the listed branch does not apply, the rule densifies via $V := \mathrm{BE.todense}$ or promotes to BCOO; those fall-back paths are not enumerated here.

Operation	Result
Shape-preserving unaries	
$\Phi(\mathrm{BE})$[3]	$\mathrm{BE}[\Phi(V),\, C,\, s,\, e,\, T]$
`cumsum`(BE)	`cumsum`(BE.todense) (no structural path)
Shape-affecting unaries	
`pad`(BE, before, after), no interior	$\mathrm{BE}[V,\, C,\, s + \mathrm{before},\, e + \mathrm{before},\, T]$
`slice`(BE, $[\ell, h)$), unit stride on out-axis	$\mathrm{BE}[V_{[\ell:h]},\, C_{[\ell:h]},\, s - \ell,\, e - \ell,\, T]$
`squeeze`(BE), `out_size`=1 at s=0, e=1	$\mathrm{BE}[V,\, C,\, s,\, e,\, T]$ (identity)
`reshape`(BE), batched→unbatched, T=False	$\mathrm{BE}[V_{\mathrm{flat}},\, C_{\mathrm{tile}},\, 0,\, \prod b \cdot \mathtt{nrows},\, T]$
`reshape`(BE), unbatched, leading singleton	$\mathrm{BE}[V_{\mathrm{reshape}},\, C_{\mathrm{tile}},\, s,\, e,\, T],\, b'{=}(1, \ldots, 1)$
`broadcast_in_dim`(BE), 1-row T=True tile to N	$\mathrm{BE}[V,\, \mathrm{broadcast}(C, N),\, 0,\, N,\, \mathrm{True}]$
`broadcast_in_dim`(BE), leading-batch insert	$\mathrm{BE}[V_{\mathrm{bcast}},\, C_{\mathrm{bcast}},\, s,\, e,\, T],\, b'{=}(\mathrm{new_batch}, b)$
`reduce_sum`(BE), axis=out, static C	$\mathrm{BE}[V_{\Sigma},\, \mathrm{dedup}(C),\, 0,\, 1,\, T]$
`reduce_sum`(BE), axis⊆batch	$\mathrm{BE}[V.\mathrm{sum}(\mathrm{axes}),\, C,\, s,\, e,\, T],\, b'{=}b \setminus \mathrm{axes}$
Indexing	
`transpose`(BE, perm), 2×2 cross-V swap	$\mathrm{BE}[V,\, C,\, s,\, e,\, \neg T]$
`transpose`(BE, perm), batch-only / V-fixed	$\mathrm{BE}[V_{\mathrm{perm}},\, C_{\mathrm{perm}},\, s,\, e,\, T]$
`rev`(BE), axis=out, unbatched	$\mathrm{BE}[\mathtt{flip}(V),\, \mathtt{flip}(C),\, N{-}e,\, N{-}s,\, T]$

[3] Φ stands for any element-wise unary that preserves shape and the zero element: $\Phi \in \{\mathtt{neg}, \mathtt{copy}, \mathtt{conj}, \mathtt{real}, \mathtt{imag}, \mathtt{convert_element_type}\}$. Each maps $V \mapsto \Phi(V)$ pointwise on the values tensor, leaving C, s, e, T untouched.

Operation	Result
`gather`$(\mathrm{BE}, \dots)$	raises `NotImplementedError` (densify upstream)
`scatter_add`$(\cdot, \mathrm{BE}, \mathrm{idx})$, kept form	`scatter_add_op`$(V, \mathrm{idx}, \dots)$ via dispatch
Shape-preserving multilinear	
$\alpha \cdot \mathrm{BE}$, scalar α	$\mathrm{BE}[\alpha\, V,\ C,\ s,\ e,\ T]$
$\mathbf{w} \odot \mathrm{BE}$, per-row weight broadcast	$\mathrm{BE}[\mathbf{w}\, V,\ \mathrm{broadcast}(C),\ s,\ e,\ T]$
BE/α, scalar α	$\mathrm{BE}[V/\alpha,\ C,\ s,\ e,\ T]$
$\mathrm{BE}_1 + \mathrm{BE}_2$, identical (s, e, b, C)	$\mathrm{BE}[V_1 + V_2,\ C,\ s,\ e,\ T]$
$\mathrm{BE}_1 + \mathrm{BE}_2$, identical (s, e, b), mixed C	$\mathrm{dedup}(\mathrm{BE}[V_1 \parallel V_2,\ C_1 \parallel C_2,\ s,\ e,\ T])$
$\mathrm{BE}_1 + \mathrm{BE}_2$, row-disjoint, abutting	$\mathrm{BE}[V_1 \parallel V_2,\ C_1 \parallel C_2,\ \min(s_i),\ \max(e_i),\ T]$
$\mathrm{BE} + \mathrm{BCOO}$ (or any incompatible)	$\mathrm{BCOO}(\mathrm{BE}) + \mathrm{BCOO}$ via shared concat path
$\mathrm{BE} - \mathrm{BE}'$	$\mathrm{BE} + (-\mathrm{BE}')$ via the add path
Shape-affecting multilinear	
$\mathrm{BE} \cdot M$, M closure 2-D, contract on in-axis	dense V_{contract} of shape (b, N, B) where $B{=}M.\mathrm{shape}[1]$
`concatenate`$(\mathrm{BE}_1, \mathrm{BE}_2)$, axis$< n_{\mathrm{batch}}$	$\mathrm{BE}[V_1 \parallel_{\mathrm{ax}} V_2,\ C_{\mathrm{widen}},\ s,\ e,\ T]$
`concatenate`$(\mathrm{BE}_1, \mathrm{BE}_2)$, axis=out	$\mathrm{BE}[V_1 \parallel V_2,\ C_1 \parallel C_2,\ 0,\ N_1 + N_2,\ T]$
Control flow / selection	
`cond`$(\mathrm{pred}, \dots)$, pred static	recurse into chosen branch jaxpr
`jit`(BE)	recurse into inner jaxpr; outputs marked traced
`select_n`$(p,\ \mathrm{BE},\ \mathrm{closures})$, single traced	$\mathrm{BE}[\mathrm{mask}_p \cdot V,\ C,\ s,\ e,\ T]$
`select_n`$(p,\ \mathrm{BE}_1,\ \mathrm{BE}_2)$, identical (s, e, b, C)	$\mathrm{BE}[\texttt{select_n}(p, V_1, V_2),\ C,\ s,\ e,\ T]$
Multi-output unary	
`split`(BE), batched, axis=out	list of $\mathrm{BE}[V_{[c_i]},\ C_{[c_i]},\ 0,\ e_i{-}s_i,\ T]$
`split`(BE), unbatched, static C	list mixing BE chunks and empty BCOO chunks

The rules above cover the BEllpack-preserving paths in `lineaxpr/_linops/ellpack_transforms.py`, `lineaxpr/_linops/ellpack.py`, and `lineaxpr/_rules/{add,mul,multilinear,structural,control_flow}.py`. Several primitives (`reshape`, `broadcast_in_dim`, `reduce_sum`) carry additional structural branches for less common shape combinations; we have shown the canonical case for each and refer the reader to the cited files for the full case-split.

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